Growing Intelligence

CounterFact Comparison Report

meta-llama/Llama-3.1-8B · 1,000 CounterFact edits · ROME / MEMIT / AlphaEdit (EasyEdit) vs Probe56 · one harness

Report built 2026-09-07 13:50 UTC · every number recounted from the raw per-row result files · growing-intelligence.com

1. EXECUTIVE SUMMARY

On the same 1,000 CounterFact edits, scored by one harness: **ROME** 0.7% success, -33.12 pp on 5,209 held-out questions, 492 of the 492 known controls broken. **Probe56** (854 hooks kept after census, on the same 1,000 targets, weights unmodified): 84.1% success, **+0.00 pp** held-out change, 0 controls broken. MEMIT and AlphaEdit: MEMIT 91.6% success / -0.10 pp, AlphaEdit 99.0% success / -1.96 pp. General benchmarks: MEMIT -0.10, AlphaEdit -1.96. Nearby facts (492 the model knew): MEMIT broke 48, AlphaEdit broke 202, ROME 492, Probe56 0. ROME collateral by edit count: 1 edit +0.00 pp, controls above floor 15; 10 edits -1.94 pp, controls above floor 163, success 10/10; 1,000 edits -33.12 pp.

2. TABLE 1 — SUCCESS AND HELD-OUT CHANGE PER METHOD

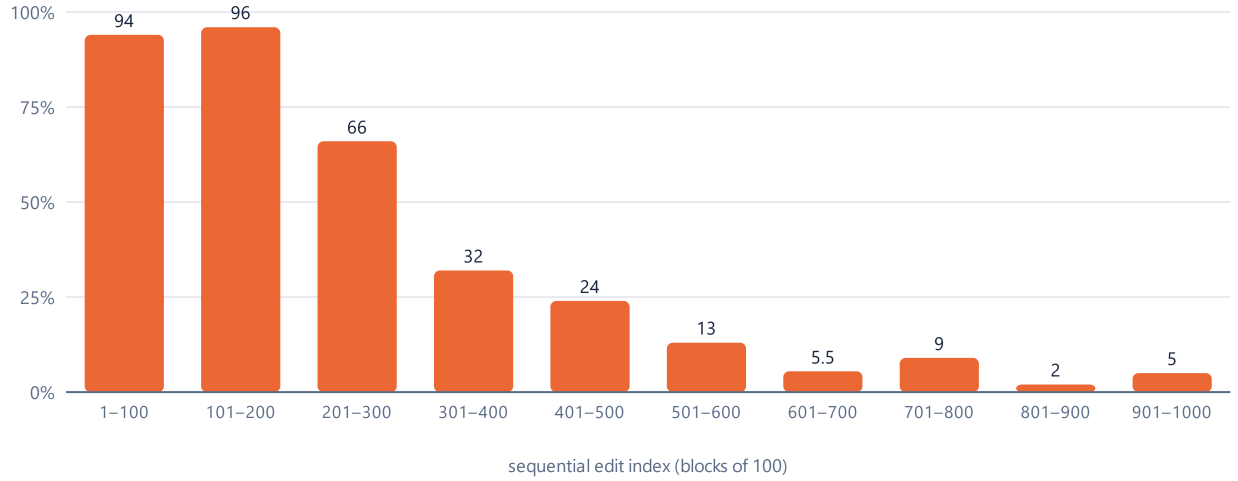
Method	Edits	Success (rank-1)	ΔMMLU-Pro	ΔGSM8K	ΔARC-Challenge	ΔTruthfulQA	ΔGPQA-Diamond	Δtotal	Controls broken (of 500, floor 8)
Probe56	1,000 targets	841/1,000 (84.1%)	+0.00	+0.00	+0.00	+0.00	+0.00	+0.00	0
ROME	1,000 edits	7/1,000 (0.7%)	-33.79	-31.82	-26.60	-35.56	-30.61	-33.12	492
MEMIT	1,000 edits	916/1,000 (91.6%)	+0.15	-1.14	-0.77	-1.85	+2.04	-0.10	48
AlphaEdit	1,000 edits	990/1,000 (99.0%)	-2.17	-1.59	+0.00	-1.85	-3.06	-1.96	202
Base (unmodified)	**0**	**2/1,000**	**33.79%**	**33.18%**	**32.23%**	**35.56%**	**30.61%**	**33.65%**	**floor 8**

Probe56 hook count (canonical, used everywhere in this report): 854 hooks kept after the collateral census, of 998 vectors fitted on the 1,000 targets; census verdict 0 above floor. Rows without a kept hook stay untouched.

Success = edit prompts whose new target is rank 1 over the full vocabulary after editing. Δ = held-out accuracy in percentage points minus the unmodified model measured once in the same regime (5,209 questions: MMLU-Pro 4,010, GSM8K 440, ARC-Challenge 391, TruthfulQA 270, GPQA-Diamond 98). Controls broken = controls at rank 1 on the unmodified model that are no longer rank 1.

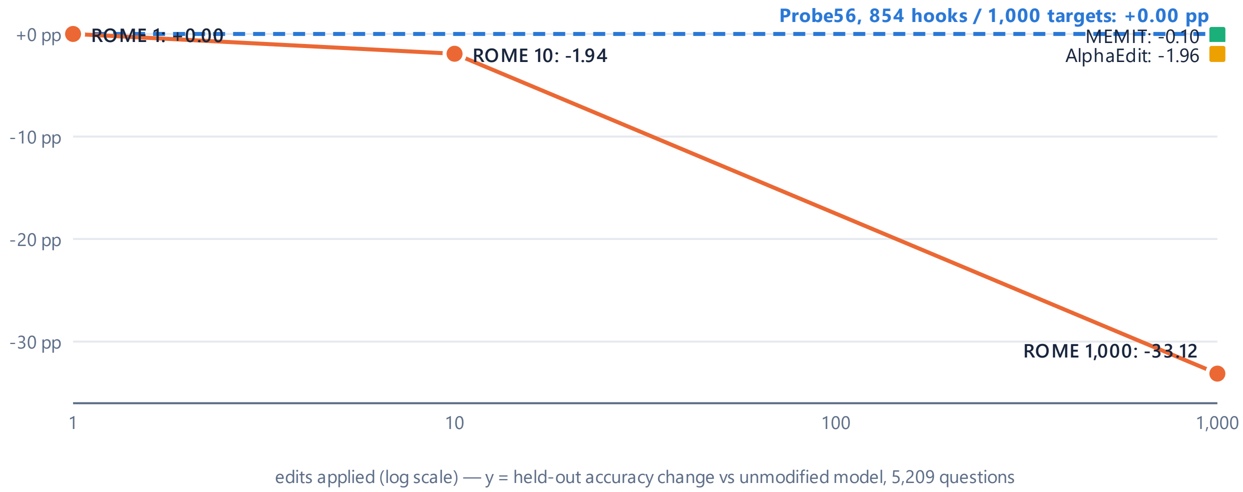
3. FIGURES

Figure 1 — ROME collapse curve



ROME works while the model is healthy; collapse begins at edit ~200. EasyEdit rewrite accuracy per 100-edit block, scored immediately after each edit: 94, 96, 66, 32, 24, 13, 5.5, 9, 2, 5. Mean 34.65% (346 of 1,000 edits at 1.0).

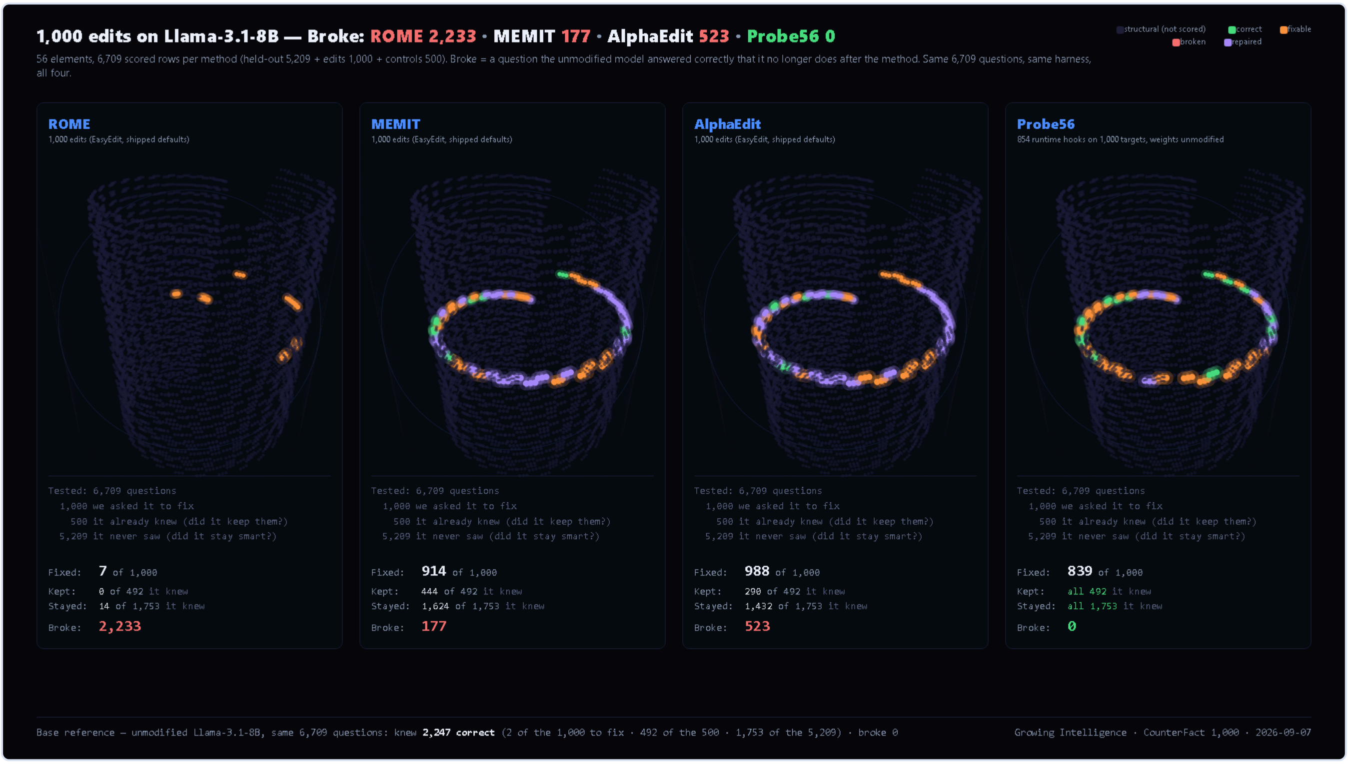
Figure 2 — ROME collateral vs edit count



ROME collateral is negligible at 1 and 10 edits and total at 1,000. Held-out change (Δtotal, 5,209 questions) after 1, 10 and 1,000 sequential ROME edits, same setup and harness. Reference line: Probe56 (854 hooks, 1,000 targets) at +0.00 pp. MEMIT and AlphaEdit plotted at 1,000 edits.

Figure 3 — Model MRI, four methods side by side

Layer-16 activation capture inside the same harness pass, 6,709 scored rows per method (held-out 5,209 + edits 1,000 + controls 500), 56 elements. Green = correct, orange = fixable (not rank 1 before or after), violet = repaired (rank 1 now, not before), red = broken (rank 1 before, not now). No extra scan; no simulation.



4. METHODOLOGY

Edits. The same 1,000 CounterFact cases for every method (seed-42 order), 500 disjoint CounterFact controls (gold = the true target), 5,209 held-out benchmark questions. ROME: 1,000 single edits applied sequentially. MEMIT: one mass edit of 1,000. AlphaEdit: sequential batches of 100. Probe56: runtime hooks on unmodified weights, fitted per question on the same 1,000 targets; 854 hooks kept after the collateral census.

Harness. One script for all four (cf_harness.py, transformers 4.55.2): rank-1 of the gold token over the full vocabulary, bf16 backbone, fp32 lm_head, left padding, padding-aware position ids, batch 24, fixed order. Edited weights were saved by EasyEdit (transformers 5.5.4) and loaded by the harness with the base tokenizer; the load was cross-checked against EasyEdit's native environment on 500/500 controls (identical ranks for every model).

ROME dose series. 1 edit: +0.00 pp, controls above floor 15. 10 edits: -1.94 pp, controls above floor 163, success 10/10. 1,000 edits: -33.12 pp, 492 controls broken.

5. STANDING ON THEIR SHOULDERS

This comparison exists because of four groups who built the field and opened their work:

Meng, Bau, Andonian, Belinkov — ROME (NeurIPS 2022) and MEMIT (ICLR 2023). They showed factual knowledge can be located and edited. They created CounterFact. Every method here, including ours, is measured on their benchmark.
https://arxiv.org/abs/2202.05262 · https://arxiv.org/abs/2210.07229

Fang, Jiang, Wang, Ma, Jie, Wang, He, Chua — AlphaEdit (ICLR 2025 Outstanding Paper). Null-space projection is elegant, and it works: 990/1,000 edits landed, -1.96 pp on general benchmarks. It is the strongest weight-editing method we measured.
https://arxiv.org/abs/2410.02355

Gu et al. (EMNLP 2024), Yang et al. (ACL 2024) — They warned that editing harms general abilities. They were right.
https://arxiv.org/abs/2401.04700 · https://arxiv.org/abs/2402.09656

EasyEdit team (Zhang et al.) — One framework, all methods, reproducible. We ran their editing code unchanged, with the run options set explicitly per each paper's protocol (ROME sequential single edits; MEMIT one mass edit of 1,000; AlphaEdit batches of 100) and the covariance statistics computed once in their own cache format, verified against their writer.
https://github.com/zjunlp/EasyEdit

6. DISCLOSURES

ROME collapse is consistent with published findings on sequential editing: Gu et al. 2024, "Model Editing Harms General Abilities of Large Language Models" (arXiv:2401.04700, EMNLP 2024); Yang et al. 2024a (arXiv:2402.09656, ACL Findings 2024); Yang et al. 2024b (aclanthology.org/2024.findings-emnlp.236); Fang et al. 2025, AlphaEdit (arXiv:2410.02355, ICLR 2025), sequential degradation on Llama3-8B, Fig. 4.

Comparison to AlphaEdit Table 1 (LLaMA3-8B, Counterfact): ROME Efficacy 64.40 ± 0.41 there; our EasyEdit rewrite_acc 34.65% (delta -29.75 pp). Caveats: different metric (teacher-forced token accuracy vs top-1 target_new over target_true), different N and schedule (1,000 single sequential edits vs 2,000 in steps of 100), Llama-3.1 vs Llama-3. Our harness criterion (rank-1 of the gold token over the full vocabulary) is stricter than either.

batch_tokens capped at 8,192 on the EasyEdit covariance pass (EasyEdit default = full 131,072 context). Covariance is a token sum; result identical. Applied to ROME, MEMIT, AlphaEdit equally (ROME's shipped config has no covariance pass, so the cap changes nothing there).

All baselines use EasyEdit code (commit 14cea8245f06) and its shipped Llama-3-8B hyperparameters, no tuning, no handicap. ROME rewrite_acc is EasyEdit's own per-edit metric, scored immediately after each edit.

One harness for all methods: rank-1 of the gold token over the full vocabulary, bf16 backbone, fp32 lm_head, left padding with padding-aware position ids, batch 24, fixed order. The unmodified model was measured once in the same regime (floor: 8 of 500 controls not rank-1).

Probe56: runtime hooks, unmodified weights, fitted per question; 998 vectors fitted, 854 hooks kept after the collateral census (0 above floor on 500 controls). 854 is the single hook count used in Table 1, the summary, the figures and the MRI panel.

MEMIT: one mass edit of 1,000 (batch_size=1,000). AlphaEdit: 10 sequential batches of 100 (batch_size=100). ROME: 1,000 sequential single edits. Each per its paper's protocol. EasyEdit defaults to batch_size=1; set explicitly.

Covariance statistics for layers 4-8 computed in a single pass, alone on the GPU, 100,000 Wikipedia samples, 68.6M tokens. Small-sample verification against EasyEdit's native writer: bit-identical. Full-scale layer-4 comparison against a native run executed under GPU sharing: max relative difference 9.3e-4 (fp32 GEMM kernel drift). Single-pass cache used for all methods; native file retained in package.

Harness loader verified against EasyEdit native environment, 500/500 controls per edited model (all controls, identical batch composition).

Probe56 scored with the gate on its fitted tensor; collateral census re-verified, 0 above floor.

Zero-edit control (fp16 round trip). EasyEdit's shipped Llama-3-8B configs load the model in fp16; the base is published in bf16, so every baseline's weights passed through fp16 before any edit (232 of 291 tensors, 3,899,715 of 8,030,261,248 elements changed by that rounding). The unmodified base was passed through the same round trip with no edit and scored on the same three splits, sterile: edits rank-1 2/1,000 (base 2); controls broken above floor 1 of 492 known; held-out 1755/5,209 (+0.04 pp vs base; 28 regressed, 30 gained). This is the floor every baseline inherits from its own shipped config; read the ROME 1-edit and 10-edit

7. APPENDIX — REPRODUCIBILITY PACKAGES

One public package per EasyEdit run: EasyEdit commit and git diff, shipped and used hparams verbatim, the run script, edit and control IDs with the selection script, EasyEdit's own run log and per-edit metrics, SHA256 of the edited weights, the per-row results, the open scoring harness, and MANIFEST.json + SHA256SUMS with the SHA256 of every file.

Run	EasyEdit commit	Files	Edited model.safetensors SHA256	MANIFEST.json SHA256	Built
ROME	14cea8245f06	38	9a528670a848d470f198cb00…	2f5472568057b7977f232771…	2026-09-07
ROME10	14cea8245f06	37	b314d2d010dc765470705787…	58e89fb6c12751ae1abbc206…	2026-09-07
ROME1	14cea8245f06	37	3a8050fcc30da10742aaca07…	ad42a39c9240d6cddfd9bd26…	2026-09-07
MEMIT	14cea8245f06	37	5686b806daf06f9d084feacd…	1bac449067af1a58f5c2819e…	2026-09-07
AlphaEdit	14cea8245f06	37	9f31a5e1f4d6eaaff43b6cb9…	c1ea6d16af95ee1ede7b8ae3…	2026-09-07

Clean-room tests of the Probe56 package (fresh pod, README only)

Test A — reproduce. fresh RunPod container, NVIDIA H100 80GB HBM3, 595.91.07 · "Ubuntu 24.04.3 LTS"; base model downloaded on the pod and hash-verified; one command from the README. Result: **PASS** — edits rank-1 841/1,000; controls known on that GPU 493, kept 493, broken 0; held-out identical to the unmodified model (0 rows differ); payload active on 0 held-out rows. Independent clean-room run (W6, H100 HBM3, pre-placed model): 841 / 493 kept, 0 broken / held-out identical / 15 min. **PASS**. Operator verification run, 2026-09-07, H100 HBM3: edits 841 / controls 493 kept, 0 broken / held-out identical / active 0 / RUN_PROBE56 **PASS**.

Test B — secrecy. Attempts on the package alone: container header field inspection; payload key unwrap with guessable keys; ciphertext entropy; printable-string scan of the executable and of its unpacked compiled module for layer, tensor, gate and vector names; open-harness scan. Result: **PASS** (ciphertext entropy 7.9999 bits/byte; no method-revealing string). Stated limit: a debugger attached to the running process can read the payload from memory.

Probe56 reproducibility package (payload encrypted and signed; hashes published)

```
container_sha256: 2dc793c5c44b93dcdf4ee12f8f35c9d6219ff89124909502f3893a555be79234
```

```
harness_sha256: 147c523613a466b10cc5f97d2eb21df3fa56771b20ec9a0b845dc784a8e6a611
```

```
manifest_sha256: 081e72ed175b23f7b3444034db13a3e860576eb25f2bf10ba74657db528ea13b
```

```
runner_bytes: 4074647416
```

```
runner_sha256: 57be7bccda759595536b688dcbe461e01b52dbe15e7728d3c2ff3bb30cb11131
```

```
sha256sums_sha256: 8c13c923dcce6d6029f679e66d84479b46f8e23c5f5cf8f3d96927c2357c52d1
```

```
tarball: aa57fd1d2465fcdf1bb60083668eebf28d30afb5c9533cf083f0a89bba1e144e
```

Raw result files (SHA256, first 16 hex) — the numbers in Table 1 are recounted from these

Method	edits	controls	held-out
Probe56	185e566179a85797	6edf8b9420320589	bdc112e79f674081
ROME	9f684dfb867e17e1	5b797c800cbd3cdc	01e3c7f1fd3ebd09
MEMIT	4966ca9461688d36	755ccc03e758f500	821a823f93fcc7c4
AlphaEdit	322a75553b379c01	5b83450c47ac9ae9	42184f3cbca21f1e
Base	0ad6d8735f836897	6edf8b9420320589	bdc112e79f674081